

mAld



Group 5

苏泳仪
陈扬靖
尹诗媛
敏敏



1. Introduction

mAld is an AI-powered cleaning assistant designed to help users manage household cleaning tasks more efficiently and with less stress. By combining AI image analysis, personalised task planning, and collaborative household features, the app guides users through the cleaning process step-by-step while reducing the mental burden of deciding where to begin. Users can upload daily room photos for AI assessment, receive targeted cleaning advice, track completed tasks, and collaborate with household members through shared cleaning plans.

Unlike traditional household chore apps that rely heavily on manual task input, mAld automatically detects common room conditions such as clutter, laundry, rubbish bins, or pet-related messes, and generates adaptive cleaning suggestions based on the room's current state. Gamification elements such as streaks, points, leaderboards, and rewards are also incorporated to improve motivation and consistency.

1.1. User Research

To better understand user pain points and expectations, preliminary user research was conducted through interviews and online discussions among students and young adults living independently or in shared spaces. The findings revealed several recurring problems:

- Many felt overwhelmed when rooms became messy and did not know where to begin cleaning.
- Shared households often experienced conflict due to uneven chore distribution or lack of accountability.
- Motivation was a major challenge, especially for repetitive cleaning tasks.

The research findings highlighted a strong demand for a solution that could reduce cognitive effort, provide personalised cleaning guidance, automate task planning, and support fair collaboration within shared households.

1.2. Purpose

The purpose of mAld is to address the common challenges people face when cleaning their living spaces, particularly:

- Feeling overwhelmed by messy environments
- Not knowing where to start cleaning
- Lack of motivation to maintain cleaning habits
- Difficulty coordinating chores within shared households



mAld aims to reduce users' cognitive load by automating planning and prioritisation, while also encouraging accountability and collaboration among household members. The app supports users in building more sustainable and manageable cleaning routines through guided assistance and progress tracking.

1.3. Scope and Objectives

mAld focuses specifically on room-level cleaning management within homes, dormitories, and shared living environments. The application integrates:

- AI-based room assessment through uploaded room images
- Step-by-step cleaning guidance and tools recommendations
- Household collaboration and task delegation
- Cleaning progress tracking and reminders
- Gamification systems such as streaks, rewards, and leaderboards

The system is designed as a mobile application that supports both individual and group cleaning management.

1.4. Assumptions

The project assumes that:

- Users are willing to upload photos for AI analysis
- Internet access is available for AI processing and cloud synchronisation
- Users prefer guided cleaning support rather than manual planning
- Shared households are open to task delegation and tracking

2. Target Users

2.1. User types and their activities

USER TYPES	ACTIVITIES
Busy Working Adults	Users with limited free time who need efficient and prioritised cleaning guidance. Activities include: ■ Uploading room photos for fast assessment ■ Receiving optimised cleaning plans ■ Scheduling tasks around work commitments ■ Tracking cleaning progress quickly



Families/Shared households	Groups living together who need fair task delegation and accountability. Activities include: ■ Assigning cleaning tasks to household members ■ Monitoring completion status ■ Collaborating through shared cleaning tasks ■ Verifying completed chores
Young Adults/Students	Users may lack cleaning experience or struggle with maintaining routines. Activities include: ■ Following step-by-step cleaning guidance ■ Learning cleaning priorities and methods ■ Building cleaning habits through gamification

2.2. User Goals

Users want an easier and less stressful way to maintain a consistently clean living environment, especially when balancing busy schedules, studies, or shared living arrangements. Many users struggle not only with the physical act of cleaning, but also with the mental burden of deciding where to start, what to prioritize, and how to divide responsibilities fairly.

Users seek:

- Clear guidance and structure so they know exactly what tasks to complete and in what order
- Personalized cleaning support tailored to their room condition and schedule
- Motivation and accountability to build long-term cleaning habits and reduce procrastination
- Fair and transparent task distribution within shared households to prevent conflict and unequal workload
- Progress tracking and visible achievements to create a sense of accomplishment and consistency
- Reduced mental load and decision fatigue by allowing AI to analyze the environment and generate actionable cleaning plans automatically
- Practical assistance and recommendations, such as suitable cleaning tools, reminders, and step-by-step instructions for inexperienced users



Ultimately, users want cleaning to feel more manageable, organized, and rewarding rather than overwhelming, repetitive, or conflict-inducing.

2.3. User modelling: Personas

PERSONAS	PRIMARY GOALS	PAIN POINTS
Name: Amanda Age: 20 Occupation: University Student Background: Lives in a dormitory alone and struggles to balance academics, social life, and room maintenance	■ Keep her room clean without spending excessive time ■ Know what to prioritise during cleaning and how to clean ■ Stay motivated to clean regularly	■ First time living independently ■ Feels overwhelmed by clutter ■ Frequently procrastinates cleaning
Name: Ryan Age: 28 Occupation: Working professional Background: Busy work schedule, shares an apartment with three roommates and often experiences disagreements over chores	■ Ensure fair division of household tasks ■ Track completed chores transparently ■ Reduce arguments about chores responsibility	■ Roommates forget assigned tasks ■ Cleaning responsibilities are uneven ■ Difficult to verify task completion

3. Business goals and Success metrics

mAld is designed for individuals living independently for the first time, as well as busy users who feel overwhelmed by cleaning tasks or unsure where to begin. Instead of simply providing cleaning tips, the app uses AI image analysis to organise cleaning into manageable, step-by-step actions. Many people delay cleaning because the process feels overwhelming or they do not know where to start.

By acting as a personal AI cleaning assistant, mAld helps users overcome this initial mental barrier and approach cleaning in a more structured and manageable way.

Business goals include:

1. Reduce cleaning paralysis through AI-guided step-by-step cleaning assistance
2. Support first-time independent living with accessible household management guidance



3. Differentiate mAld through personalised AI image analysis and tailored cleaning recommendations

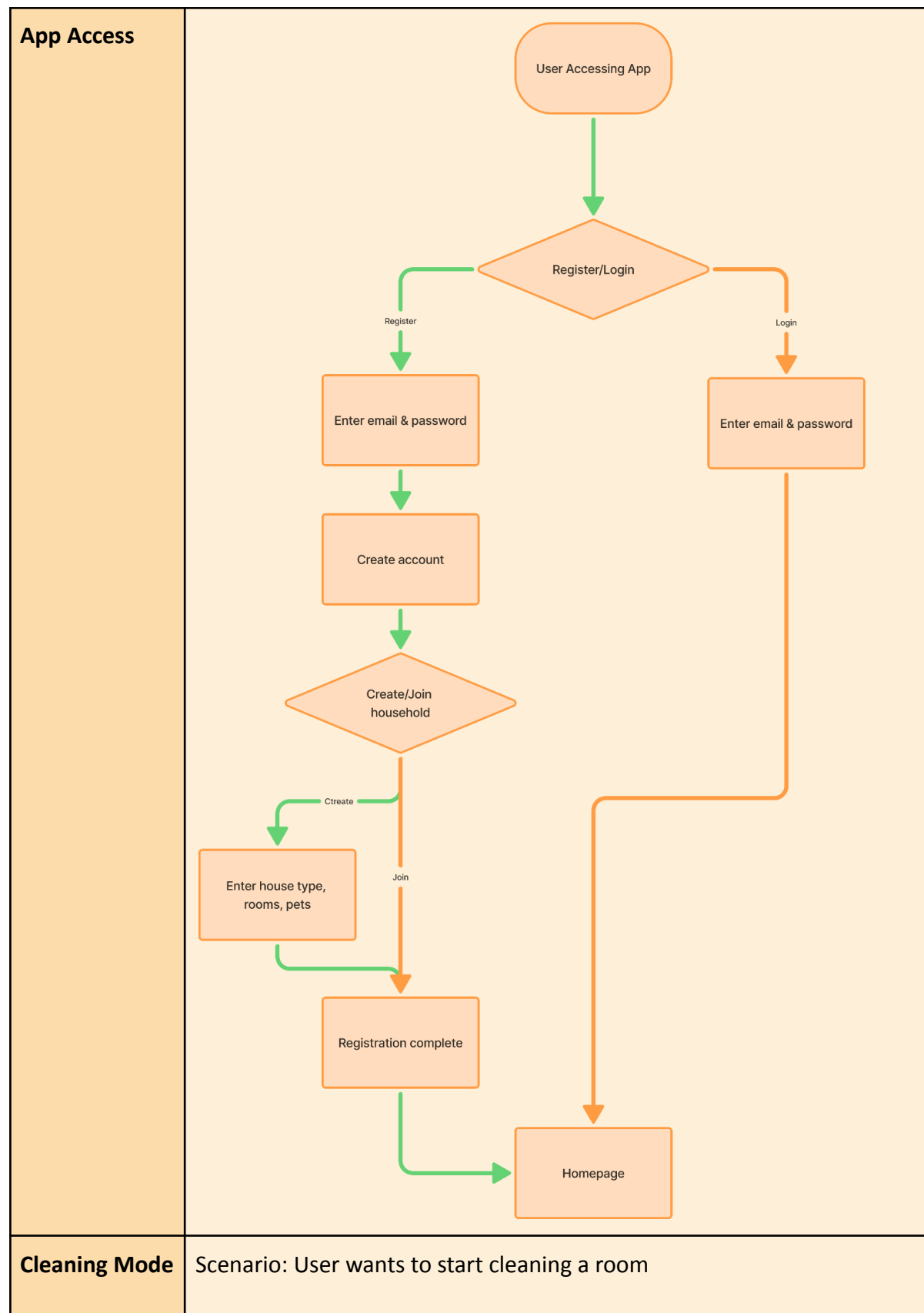
SUCCESS METRICS	
Number of app downloads	Measures initial interest and awareness of the app
User sign-up rate	Tracks how many downloaded users create accounts and begin using the platform
Monthly Active Users	Measures how many users consistently return to the app over time
Cleaning task completion rate	Tracks if users complete the AI-generated cleaning steps, evaluates whether the guidance was practical and effective
Retention rate	Tracks whether users continue using the app after their first experience
Reduction in cleaning overwhelm	Measured through survey or feedback prompts
Improvement in cleaning consistency	Tracks whether users clean more regularly after using the app (i.e. week streak)
AI recommendation rating	Measures if users feel the AI correctly prioritised cleaning tasks
Reward redemption rate	How often users redeem points for cleaning-related rewards, measure whether incentives successfully motivate engagement
Partnership growth	Measures the number of brands and companies collaborating with the app

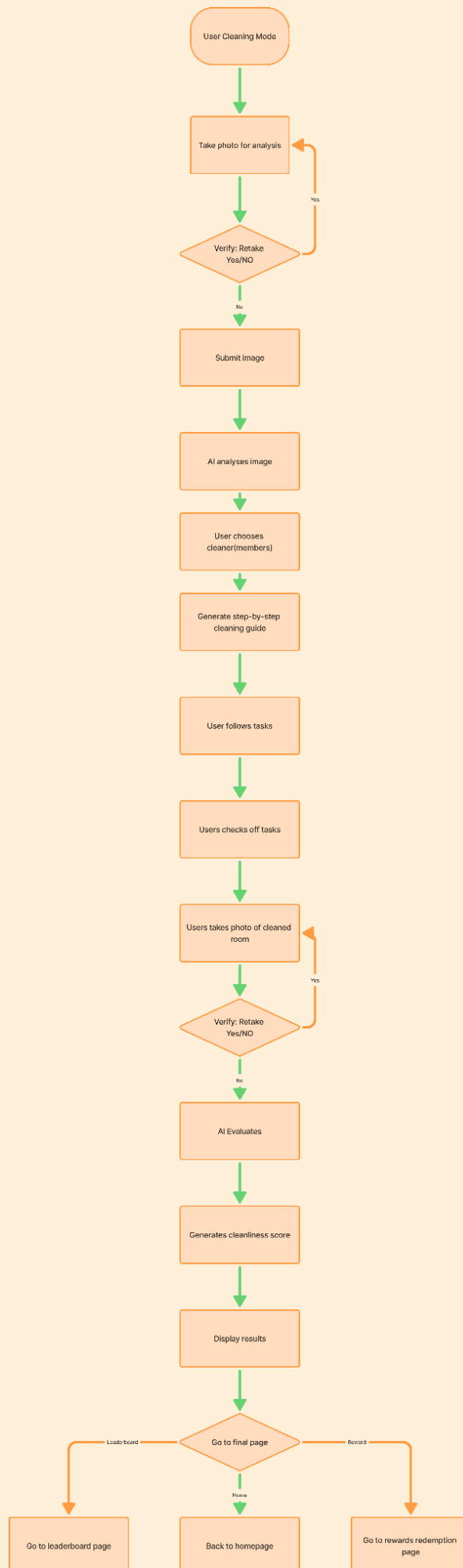
4. Task Analysis

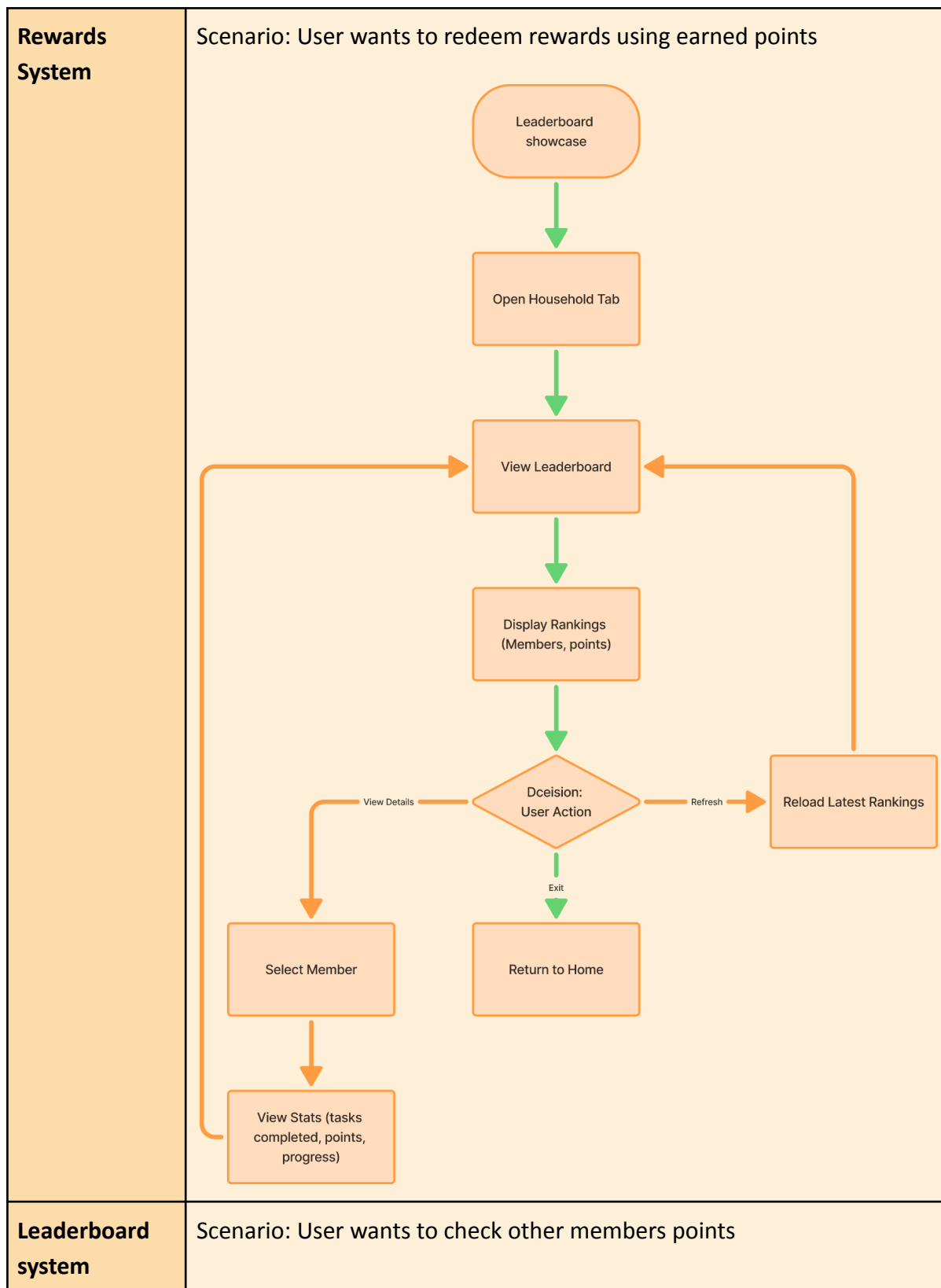
4.1. Modules

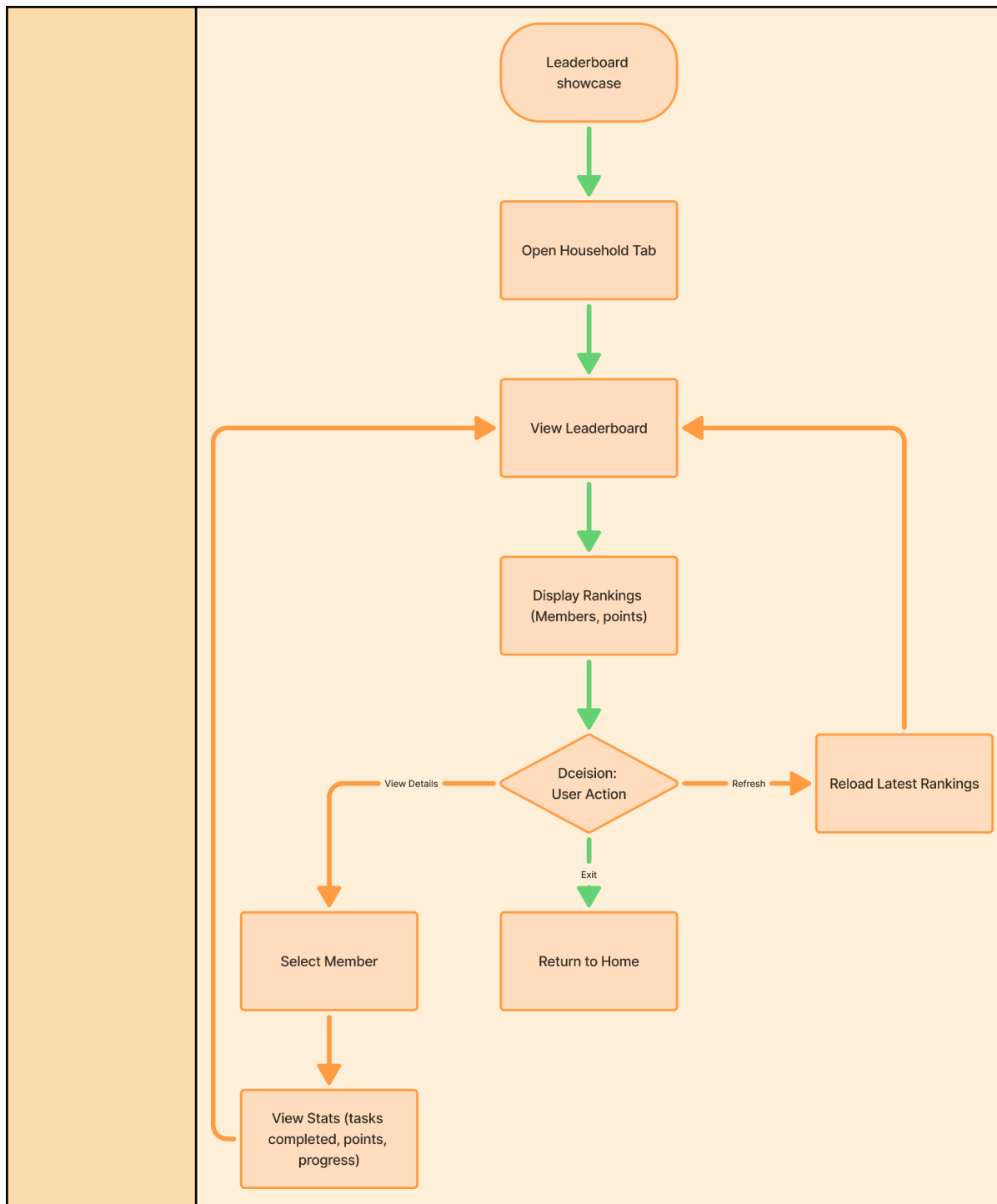
TASK ANALYSIS	
Modules	Scenario: User wants to register/login to access the app
	WORKFLOW











5. AI Component Analysis

5.1. Data requirements

DATA TO BE STORED	
Account & Profile	■ Login info ■ User Preferences ■ Reward Points



Household system	■ Household members ■ Roles ■ Leaderboard scores ■ Stats
Cleaning activity	■ Before/after images (Camera feature) ■ Tasks assigned (delegation) ■ Completion status
AI guide + results	■ Detected objects (mess, clutter types) ■ Suggested cleaning steps ■ Cleaning scores / Progress
Rewards system	■ Points earned ■ Redeemed rewards ■ Rewards list

GENERATED DATA	
AI generates	■ “Mess detected” labels (e.g. clothes, trash, table mess) ■ Tasks Sequences (“Clean table” -> “Sweep the floor”) ■ Cleanliness score ■ Efficient cleaning room
Users generate	■ Tasks completion data ■ Before/after improvements ■ Engagement (leaderboards, rewards)

HOW THIS POWERS DATA FLYWHEEL
1. More users take cleaning photos AI learns different room types and mess patterns
2. Users follow tasks efficiently/inefficiently System learns how to optimise recommendations
3. Before/after images AI improves understanding of what a “clean” room looks like



4. Rewards + leaderboards Increases engagement between household members -> More data generated

DATA STORAGE	
Images	Cloud storage (before/after photos, profile picture)
App Data	Database (Users, households, rewards, tasks)
AI Data	Logs + labels stored for retraining

5.2. Model requirements

MODEL REQUIREMENTS	
Type of model	Computer vision based AI model including: ■ Object detection component (e.g. Detects items like clothes and trash) ■ Image classification component (rating room cleanliness level) ■ Vision and language component to generate cleaning instructions
Input and Output	<i>Input:</i> User-uploaded room images <i>Processing:</i> Detects objects and clutter level <i>Output:</i> Identified messy areas, prioritised cleaning steps, cleanliness score
Core functions	■ Detects messy objects (clothes, trash, dishes) ■ Identify areas (bed, floor, desk) ■ Generate cleaning steps (what to clean first and where to start) ■ Evaluate cleanliness (before VS after) ■ Assign cleanliness score
Training Approach	Supervised learning using labelled room image datasets <i>Training data includes:</i> ■ Annotated objects (e.g. clothes, trash, waste) ■ Cleanliness levels



	<i>Continuous retraining using:</i> ■ User-uploaded images ■ Before and after comparison ■ User interaction data
Performance needs	■ High accuracy in detecting clutter ■ Fast response (near real-time after photo taken) ■ Works under different lighting / room conditions
System constraints	■ Must run efficiently on mobile/cloud ■ Privacy (user images/rooms handled securely)

5.3. Analysis of Human-AI Interaction logic

The system primarily follows a cognitive design approach, where the AI takes the lead in guiding users through the cleaning process. By analysing the user-uploaded room images, the AI breaks down the complex cleaning tasks into clear, structured, step-by-step instructions. This reduces users' cognitive load and prevents them from feeling overwhelmed when faced with messy environments, allowing them to focus on executing tasks rather than planning what to do next.

In addition, the system incorporates effective design elements to enhance user motivation and engagement. Features such as rewards, progress tracking and household leaderboards create a sense of achievement and satisfaction, encouraging users to continue using the application. This is further supported by persuasive design principles, here the AI guides toward efficient cleaning behaviours through recommended tasks sequences and gamified elements, promoting consistent and effective cleaning habits over time.

Overall, the system demonstrates a strong human-AI collaboration model where the AI provides structured and efficient guidance while maintaining user engagement through motivational and behavioural design strategies.

5.4. Data Flywheel

- User takes room photos
- Photos allow AI to understand more unique rooms type/clutter
- AI detects clutter & suggests cleaning steps
- User completes cleaning
- System collects results (images + task performance)
- Data used to improve and teach AI model
- Better cleaning recommendations and detection



- Improved cleaning efficiency and style attracts new users, allowing for more datasets
- (loops back)

The data flywheel describes how the system continuously improves its performance through user interaction. When users take photos of their rooms, the AI analyses the images to detect clutter and generate cleaning instructions. After users complete the cleaning tasks, the system collects important data such as before-and-after images and task execution performance (e.g. time taken and completion results).

This data is then fed back into the system to improve the AI model. Over time, the AI becomes better at recognising different types of mess and generating more effective and accurate cleaning recommendations. As the system improves, users achieve better cleaning outcomes, which encourages continued usage and generates even more data.

Overall, this creates a continuous feedback loop where increased usage leads to better AI performance, which in turn enhances user experience and drives further engagement.

6. Constraints and Risks

While mAld aims to simplify cleaning and reduce user overwhelm through AI guidance, the platform also faces several constraints and potential risks that may affect its growth, user retention, and long-term sustainability

CONSTRAINTS	
AI image recognition limitations	AI may struggle to accurately analyse cluttered environments, poor lighting, or overlapping objects, which could affect the quality of recommendations
Limited training data	Requires large and various datasets of rooms and cleaning scenarios to provide accurate and personalised guidance
User privacy concerns	As users upload photos of personal spaces, the app must securely store images and clearly communicate privacy policies to mitigate concerns
User motivation and long-term engagement	Due to the perception of cleaning being a chore, it would be difficult to maintain long-term user engagement and retention
Partnership collaborations	mAld's reward system depends on securing and



	maintaining partnerships with cleaning-related brands. As a new platform, the app may initially struggle to attract companies to provide incentives
--	---

RISKS	
Inaccurate AI recommendations	If the AI provides unrealistic or unhelpful cleaning instructions, users may lose trust in the platform. Moreover, can lead to frustration is AI is prioritising the wrong tasks, or failing to identify key clutter areas
Privacy and security risks	A data breach involving room images or personal information could severely damage user trust and brand reputation
Over-reliance on gamification	If users engage only for rewards rather than actual habit formation, engagement may drop if incentives are reduced
AI bias	AI may not account for different cultural cleaning standards, varying lifestyles, disabilities, or different living conditions. As a result, it could make the recommendations feel less personalised or inclusive
Low user retention	Users may try the app once but fail to build long-term cleaning habits as cleaning is not typically a daily activity



7. User Interface Screen Flow

